Puri District Administration Initiative

Nirikhyana

Proof of Concept (POC)
For
Design, Development, Implementation & Maintenance of the Mobile App & Web Portal
for
Maternal & Infant Health Monitoring System (Nirikhyana)

POC. Letter No. STPL/DEV-040120261356

Version-1

Submitted by,

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Revolutionizing maternal healthcare with Nirikhyana- An innovative technology.

An AI-driven Mobile App & Web portal seamlessly connecting all stakeholders for efficient USG referrals, ensuring efficient care for pregnant women across Puri District.

Table of Content

Definitions, Acronyms, and Abbreviations

- **PW** – Pregnant women
- **USG** – Ultrasound Sonography
- **ANM -** Auxiliary Nurse and Midwife
- **HWF** – Health worker female
- **ANC -** Antenatal Care
- **HWM** – Health worker male/Supervisor
- **MMR -** Maternal/Infant Mortality Rate
- **PCPNDT -** Pre-Conception and Pre-Natal Diagnostic Techniques
- **AI** – Artificial Intelligence
- **DLT -** Distributed Ledger Technology
- **REST -** Representational State Transfer Application Programming Interface
- **RDBMS -** Relational Database Management System
- **API -** Application Programming Interface
- **RBAC -** Role-based access control

1. Introduction

A robust IT infrastructure, comprising the **Nirikhyana mobile application and dashboard** (web portal), enables seamless integration and coordinated service delivery across multiple levels of the healthcare system. The Nirikhyana application facilitates efficient referral of pregnant women requiring ultrasound examinations, while the dashboard provides a consolidated and role-based view of data from sub-centre to block and district levels, in strict compliance with the **PCPNDT Act**.

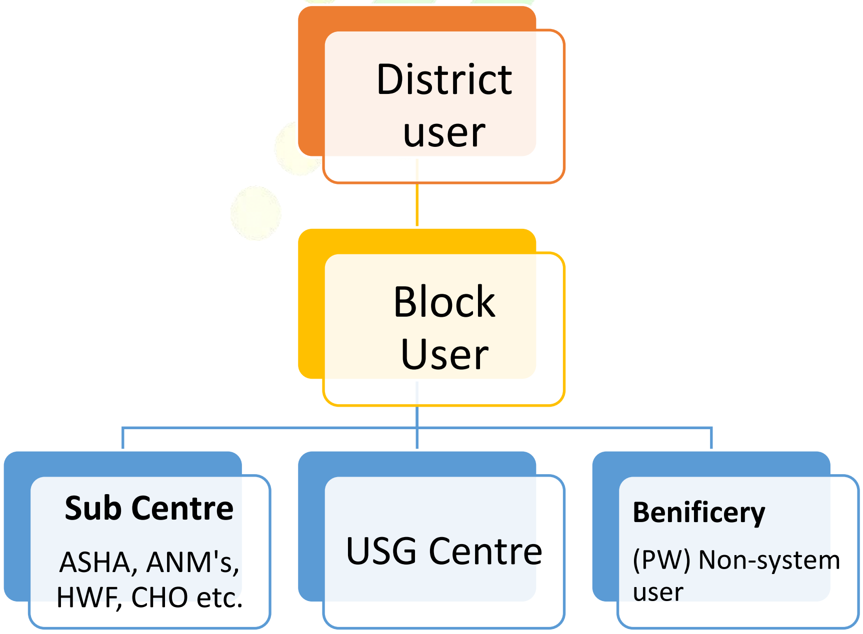
The platform incorporates **AI-enabled analytics** to support early identification of high-risk pregnancies, assist in data-driven decision-making, and improve follow-up efficiency after USG examinations. This integrated approach enhances transparency, accountability, and continuous monitoring of maternal and infant health services. By digitally connecting frontline health workers, medical officers, and ultrasound centres, the system ensures timely interventions and systematic follow-up until safe institutional delivery, ultimately contributing to improved maternal and neonatal health outcomes.

2. Vision

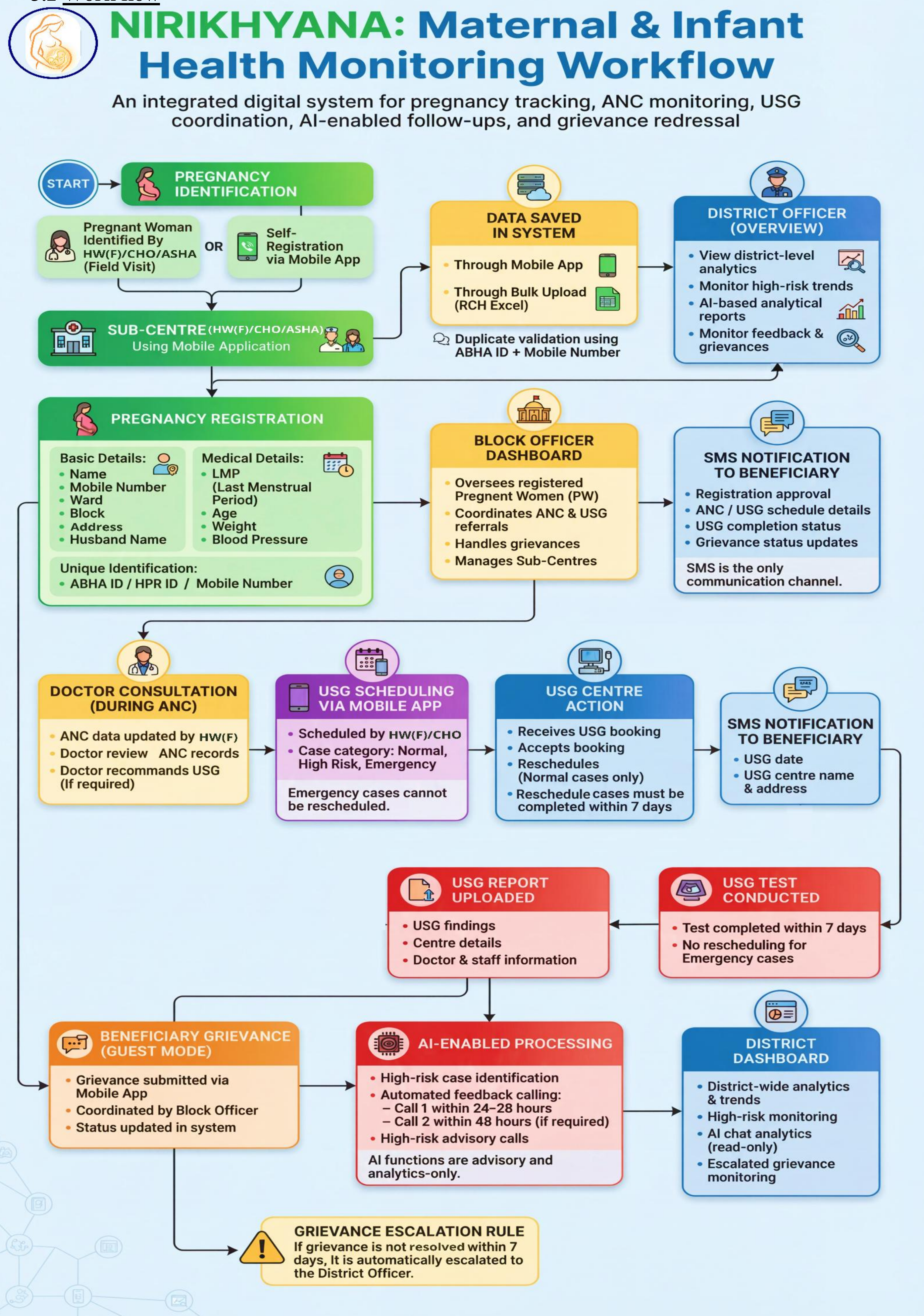
The vision of the **Nirikhyana** initiative is to establish a **decentralized yet seamlessly integrated digital ecosystem** for maternal and infant healthcare. By harnessing technology, the platform aims to bring together Auxiliary Nurse Midwives (ANMs), expectant mothers, medical practitioners, ultrasound centres, and administrative authorities on a unified system that supports coordinated and proactive care. At the core of this vision is the role of frontline health workers in initiating and guiding the maternal and infant care continuum, supported by digital tools that enable consistent tracking, monitoring, and follow-up at every stage. The system aspires to ensure that each phase of the care journey is transparent, accountable, and optimized for timely decision-making, ultimately contributing to improved health outcomes for mothers and infants.

3. Detailed Scope

3.1 User's Characteristics: Five types of users with role-specific permissions.



3.2 Work flow


NIRIKHYANA: Maternal & Infant Health Monitoring Workflow
An integrated digital system for pregnancy tracking, ANC monitoring, USG coordination, AI-enabled follow-ups, and grievance redressal
START
PREGNANCY IDENTIFICATION
Pregnant Woman Identified By HW(F)/CHO/ASHA (Field Visit)
OR
Self-Registration via Mobile App
SUB-CENTRE (HW(F)/CHO/ASHA)
Using Mobile Application
DATA SAVED IN SYSTEM
Through Mobile App
Through Bulk Upload (RCH Excel)
Duplicate validation using ABHA ID + Mobile Number
DISTRICT OFFICER (OVERVIEW)
View district-level analytics
Monitor high-risk trends
AI-based analytical reports
Monitor feedback & grievances
PREGNANCY REGISTRATION
Basic Details:
Name
Mobile Number
Ward
Block
Address
Husband Name
Medical Details:
LMP (Last Menstrual Period)
Age
Weight
Blood Pressure
Unique Identification:
ABHA ID / HPR ID / Mobile Number
BLOCK OFFICER DASHBOARD
Oversees registered Pregnent Women (PW)
Coordinates ANC & USG referrals
Handles grievances
Manages Sub-Centres
SMS NOTIFICATION TO BENEFICIARY
Registration approval
ANC / USG schedule details
USG completion status
Grievance status updates
SMS is the only communication channel.
DOCTOR CONSULTATION (DURING ANC)
ANC data updated by HW(F)
Doctor review ANC records
Doctor recommands USG (If required)
USG SCHEDULING VIA MOBILE APP
Scheduled by HW(F)/CHO
Case category: Normal, High Risk, Emergency
Emergency cases cannot be rescheduled.
USG CENTRE ACTION
Receives USG booking
Accepts booking
Reschedules (Normal cases only)
Reschedule cases must be completed within 7 days
SMS NOTIFICATION TO BENEFICIARY
USG date
USG centre name & address
USG TEST CONDUCTED
Test completed within 7 days
No rescheduling for Emergency cases
USG REPORT UPLOADED
USG findings
Centre details
Doctor & staff information
BENEFICIARY GRIEVANCE (GUEST MODE)
Grievance submitted via Mobile App
Coordinated by Block Officer
Status updated in system
AI-ENABLED PROCESSING
High-risk case identification
Automated feedback calling:
– Call 1 within 24–28 hours
– Call 2 within 48 hours (if required)
High-risk advisory calls
AI functions are advisory and analytics-only.
DISTRICT DASHBOARD
District-wide analytics & trends
High-risk monitoring
AI chat analytics (read-only)
Escalated grievance monitoring
GRIEVANCE ESCALATION RULE
If grievance is not resolved within 7 days, It is automatically escalated to the District Officer.

3.3 Technical Specification

- Development of a dynamic web portal & mobile app (Android) as per the Govt. Standards.
- Multilingual (English & Odia) based for smooth operation.
- Integration of CMS facilities into the web portal.
- Web portal will be compatible with all types of browsers like Chrome, Mozilla Firefox, Internet Explorer & all types of devices.
- Rich User Interface (Web 3.0) based on frameworks like Bootstrap and other requisite frameworks needs to be used.
- Interactive multimedia-oriented web portal.
- This supports responsive page design. Compatible with multiple-size screens (computers, tablets, mobiles, and any other electronic screen media device). The web adjusts itself automatically as per the screen resolution of the Computer or Mobile.
- The mobile app & web portal complies with all applicable Government of India cybersecurity guidelines, including CERT-In, MeitY, and GIGW 3.0. The system will undergo VA/PT and cybersecurity audit by a CERT-In empanelled agency prior to going live.
- Scrumlin shall ensure compliance with applicable Government of India data protection regulations and safeguard all citizen and health-related data for authorized government use only. All data shall remain the exclusive property of the concerned Government authority, with confidentiality maintained during and after the contract period.
- The Nirikhyana App provides a simple and intuitive interface that allows Sub-Centre users to effortlessly book consultation requests for pregnant women in their jurisdictions.
- **User-Friendly Interface:** The Nirikhyana App features a clean and intuitive interface designed to help ANMs easily register pregnant women and schedule consultation requests within their respective sub-centres, ensuring smooth and efficient data entry with minimal effort.
- **Integrated Referral System:** Following medical consultation, pregnant women can be digitally referred to designated ultrasound centres through the application, enabling a smooth, well-coordinated, and efficient referral process.
- **Real-Time Data Monitoring:** Information captured during medical consultations and ultrasound examinations is updated instantly within the application, enabling real-time visibility and supporting accurate and timely data tracking.
- **Centralized Reporting:** Ultrasound centres can upload scan findings directly into the app, enabling centralized consolidation of data and generation of unified reports for monitoring and review.
- **Multi-Level Monitoring Dashboard:** The system offers role-based dashboards for different administrative levels, enabling block authorities to monitor sub-centre activities and providing district officials with a consolidated view of performance across blocks and sub-centres to support transparency and accountability.
- **Automated Notifications:** The application automatically sends alerts and reminders to relevant stakeholders to support timely consultations and ultrasound examinations, helping minimize missed or delayed appointments.
- **In-built Analytics:** The application incorporates AI-driven analytics to support district and block-level administrators in generating actionable insights, identifying high-risk cases, and improving planning and delivery of maternal and infant healthcare services.
- **Grievance Management:** Pregnant women can raise grievances directly through the mobile application. Upon submission, the grievance is automatically notified to the Block user for coordination with the respective sub-centre and resolution. If the grievance is not resolved within seven (7) days, it is automatically escalated and reflected on the district dashboard for further action and monitoring.

3.3.1 Proposed Solution Overview

Solution Components

Sl#	Component	Description
1	Web portal Login access	District & Block
2	Mobile App (Android)	District, block, Sub-centre, USG Centre & PW self-registration

3	SMS Engine for	Alerts, reminders, advisories, tips
4	AI Auto Calling	Feedback collection in Odia
5	AI & Analytics Layer	Reports, insights, compliance monitoring
6	Secure Database	Centralized maternal health data

3.3.2 Core Components

- Web & Mobile App
- User & Role Management
- PW Registry & Bulk Excel Upload
- ANC & USG Scheduling
- Emergency Case Handling
- USG Report Upload & Verification
- Grievance Redressal System
- Connect directly to the beneficiary
- AI analytics dashboard.
- MIS reports & data analytics
- AI integration for call analytics
- Odia SMS (DLT) integration

3.3.3 Technical Architecture

Frontend ---- Responsive Web + Android Mobile App

Backend ---- REST-based Application Service

Database ---- RDBMS with Analytics Store

AI Layer ---- Reporting & auto call orchestration

Hosting ---- India-based cloud storage

3.3.4 Security Framework

- RBAC system
- Encryption at rest & transit
- Secure file uploads (Excel Reports)
- Audit trails and activity logs.

3.3.5 Data Migration & Bulk Upload

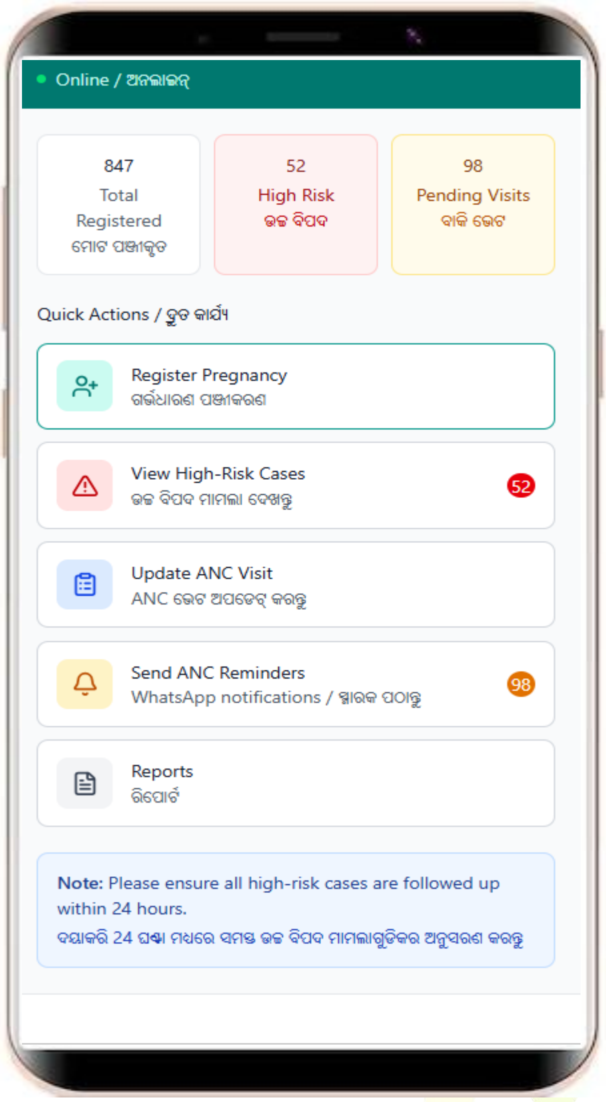
- Excel-based bulk upload (Government-defined format)
- Duplicate detection using ABHA ID + Mobile Number
- Auto-segregation to Block / Sub-centre
- Upload validation and error reporting.

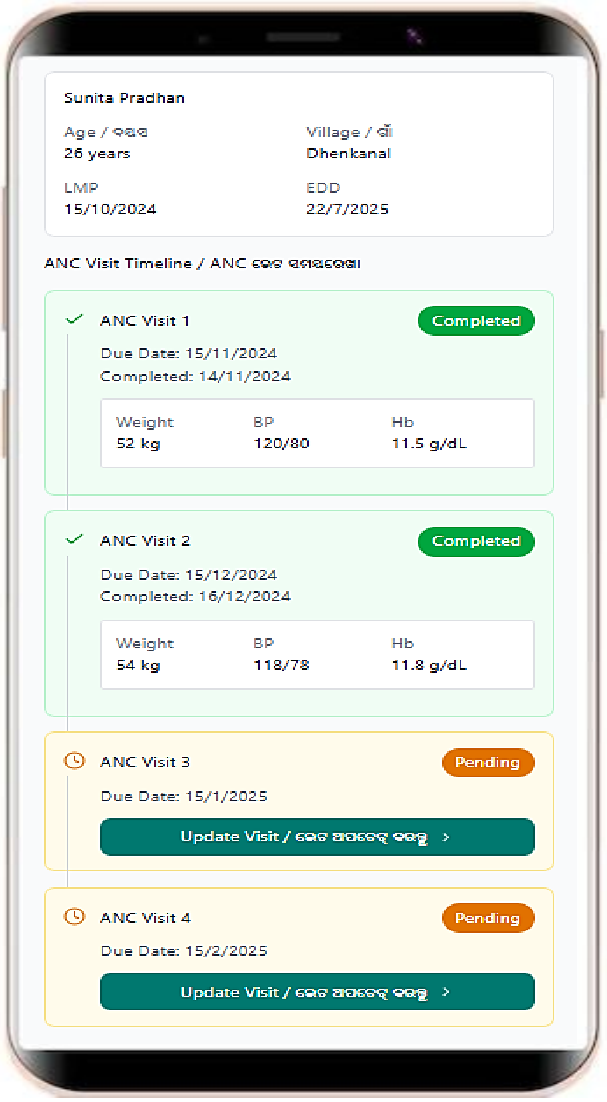
3.3.6 AI and Automation Features

- Automated calling (Odia language)
- High-risk pregnancy indicators.
- Compliance breach alerts (7-day USG rule)
- Feedback analytics for District monitoring.
- Performance dashboards for Blocks

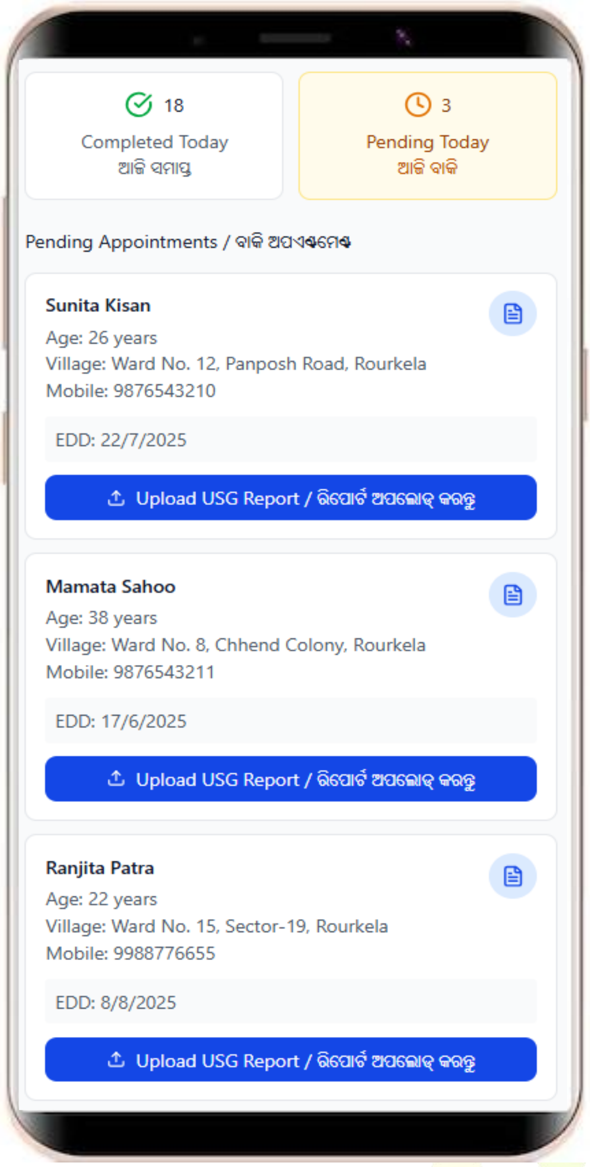
4. Prototype/Wireframe

4.1 Sub-Centre User interface

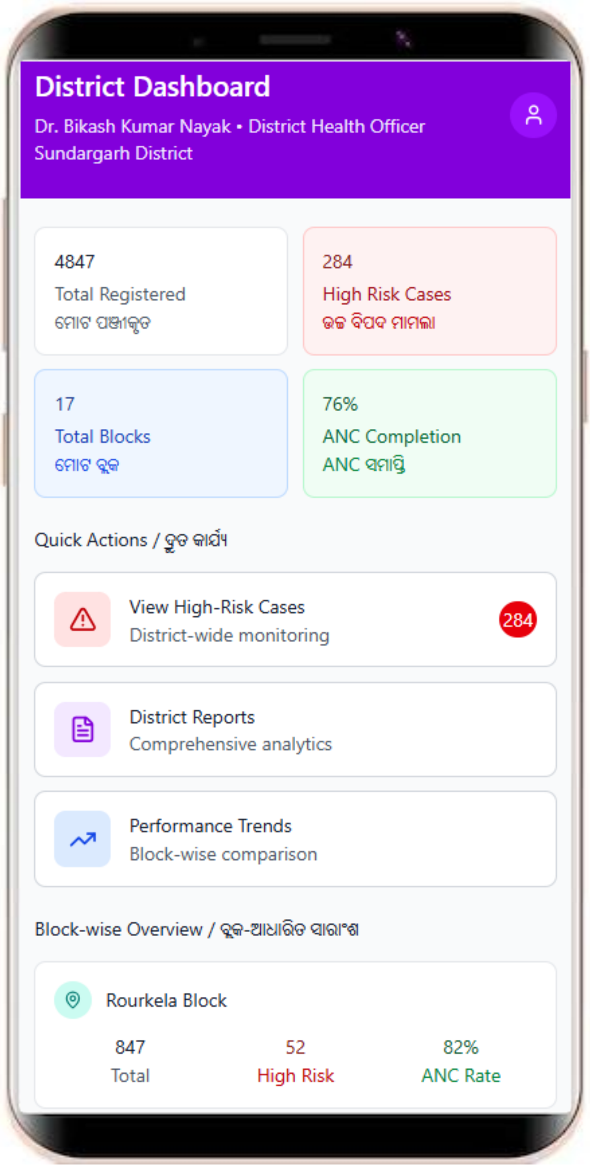




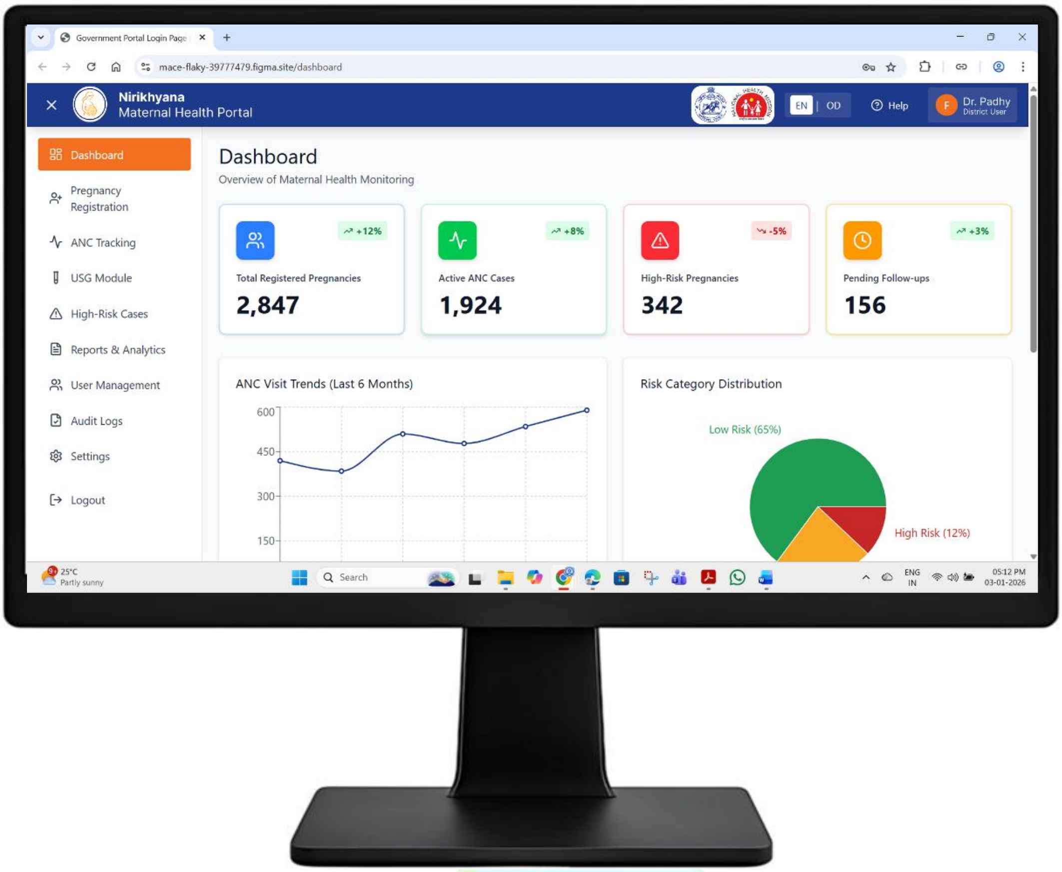
4.2 USG Centre interface

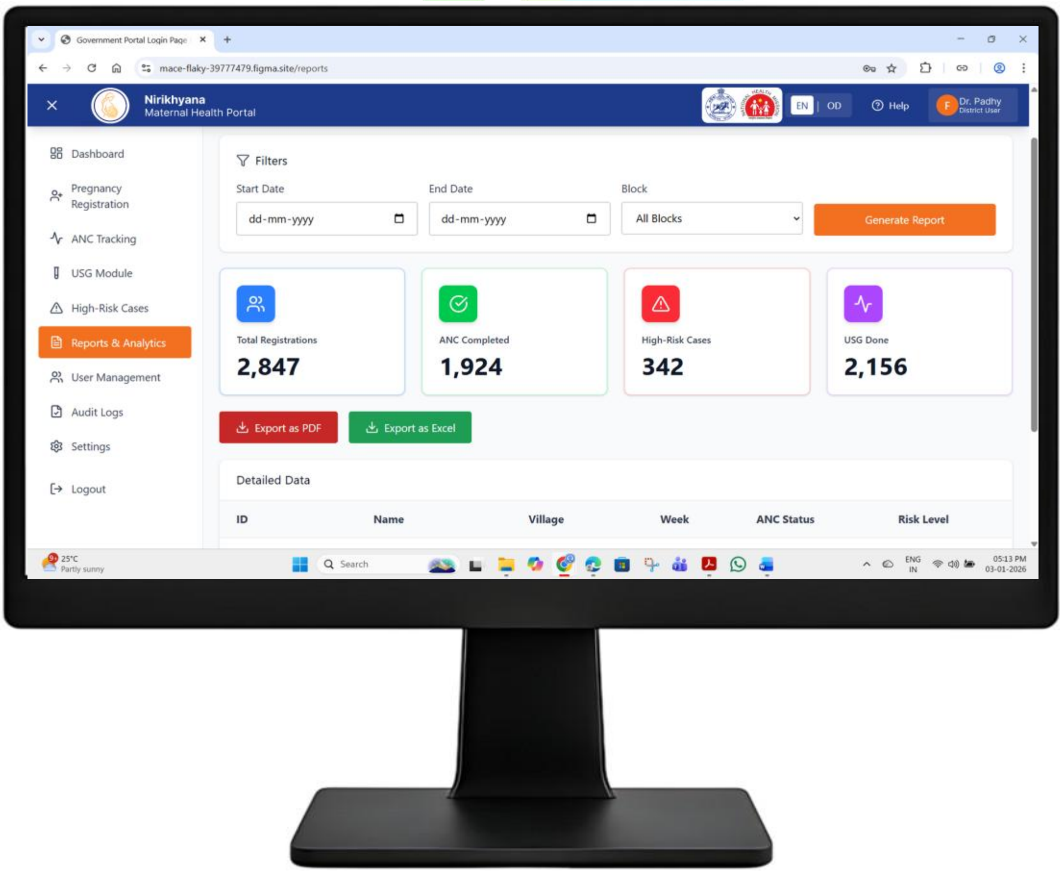


4.3 District user interface



4.4 District Dashboard & Reports





5. **Summary / Conclusion**

The **Nirikhyana Proof of Concept (POC)** establishes a structured, technology-enabled framework to support the objectives of the National Health Mission (NHM) and district maternal and infant health programs. The proposed mobile application and web portal provide a standardized digital mechanism for pregnancy registration, ANC monitoring, USG referral management, high-risk identification, grievance redressal, and administrative oversight, while ensuring alignment with statutory and programmatic requirements, including the PCPNDT Act.

The POC demonstrates how digitization of frontline service delivery processes can enhance transparency, accountability, and coordination across sub-centre, block, and district levels. Role-based dashboards and centralized reporting enable district and block authorities to monitor service coverage, compliance timelines, and high-risk trends in near real time, thereby supporting evidence-based planning and timely corrective actions. The integration of **AI-enabled analytics and automation**, used strictly as a decision-support mechanism, further strengthens monitoring efficiency without replacing clinical or administrative judgment.

The proposed system has been designed in compliance with **Government of India cybersecurity and data protection guidelines**, ensuring secure handling of sensitive maternal and health-related data. Its modular and scalable architecture allows for phased implementation and future enhancement, as required by the administering authority.

Overall, the Nirikhyana POC validates the readiness of the proposed solution for operational deployment as a district-level digital health intervention. It provides a policy-aligned, sustainable, and governance-friendly platform that can contribute to improved maternal and infant health outcomes through strengthened monitoring, timely interventions, and informed administrative decision-making.

Yours faithfully,

Authorised Signature
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